Equipment, Supplies and Clinical Support Services Management

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1. Purpose

This document explains how medical equipment, medical gases, pharmacy stock, laboratory, radiology and the blood bank are managed at CareOps General Hospital. Equipment and support service delays are a common hidden cause of long waiting times and blocked beds.

2. Critical equipment inventory

The hospital has 40 ventilators, of which 24 are in the ICU, 6 in the NICU and operating theatres, 6 in the ED and 4 in reserve. It has 25 defibrillators, 150 patient monitors, 300 infusion pumps, 2 CT scanners and 1 MRI scanner, 12 portable X-ray machines and 10 ultrasound machines. Each item has an asset tag and a status in the equipment register.

3. Equipment status codes

Available means the item is ready for use. In use means the item is attached to a patient or is in a procedure. Under maintenance means it is in planned servicing. Out of service means it has a fault and is not safe to use. Awaiting calibration means its calibration date has passed and it must not be used until calibrated. Equipment utilization is hours in use divided by hours available. Sustained utilization above 85 percent for critical items means there is little reserve capacity.

4. Maintenance

Preventive maintenance is planned servicing done on a fixed schedule. Ventilators, patient monitors and infusion pumps are serviced every 6 months. Defibrillators are tested every day by the unit nurse and serviced every 6 months. CT and MRI scanners are serviced according to the manufacturer schedule, with quality checks every month. Corrective maintenance is the repair after a fault. Calibration is done once a year for measuring devices. The Biomedical Engineer on duty responds to an urgent equipment fault within 30 minutes and a critical fault in an ICU or ED within 15 minutes. A replacement or backup unit must be provided within 1 hour for life-support equipment.

5. Daily checks

Each ED and ICU nurse in charge checks the crash cart and defibrillator at the start of every shift, confirms the seals and expiry dates, and records the check. Oxygen cylinders are checked for pressure at each shift. The suction unit and the bag-valve-mask are checked daily.

6. Equipment shortage response

If free ventilators or defibrillators fall below 20 percent of the total, the Biomedical Engineer informs the Duty Manager. If they fall below 10 percent, the Duty Manager arranges to move equipment from low-use areas, brings reserve ventilators into service, and informs the Medical Superintendent. A sudden usage spike of more than 150 percent of the normal average for the same time of day is investigated for a patient surge, a device left running or a sensor error.

7. Medical oxygen

The hospital keeps a minimum reserve of oxygen sufficient for 24 hours of average consumption. The Duty Manager is told if the reserve falls below 24 hours. The supplier is called to refill when the liquid oxygen tank falls to 40 percent. Backup cylinders are stored at the ICU, the ED and the operating theatres.

8. Pharmacy

Each item has a minimum stock level and a reorder level. Critical emergency medicines must never be out of stock. A stock-out of a critical medicine is reported to the Duty Manager within 1 hour. The pharmacy checks expiry dates monthly and removes expired medicines through the biomedical waste route. Discharge medicines are prepared within 60 minutes of the discharge order.

9. Laboratory turnaround targets

Urgent or STAT laboratory tests from the ED and ICU have a target result time of 60 minutes from sample receipt. Routine inpatient tests have a target of 4 to 6 hours. Critical values, which are results that are dangerously abnormal, are phoned to the responsible nurse or doctor within 15 minutes. A rise in laboratory turnaround time is a frequent cause of longer ED stay.

10. Radiology turnaround targets

ED X-rays are done within 30 minutes of the request. A STAT CT scan is done within 60 minutes. Routine CT and MRI scans for inpatients are done within 24 hours. STAT reports are given within 60 minutes and routine reports within 24 hours.

11. Blood bank

The blood bank keeps blood components under controlled storage. Red cell units are stored at 2 to 6 degrees Celsius in a blood bank refrigerator and can be kept for up to 42 days when preserved in an additive solution. Platelets are stored at 20 to 24 degrees Celsius with continuous gentle agitation and last about 5 days. The blood bank keeps a minimum emergency stock of O negative red cells. The blood bank informs the Duty Manager when stock of any group falls below 3 days of average use. Emergency blood issue is made within 15 minutes of a request.